Network Security & Segmentation Implementation Report

Prepared By: Bibek Pandeya
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Project: Corporate Network Virtualization, Security Segmentation, and Packet Analysis

Executive Summary

This report outlines the successful design, implementation, and security auditing of a segmented corporate network using pfSense, Ubuntu (LAN), and Metasploitable (DMZ). The objective was to take a standard Class C IP block (192.168.10.0/24) and divide it into three distinct, isolated networks (LAN, DMZ, and Guest) to enforce strict access control. Core services including DHCP, DNS, and NAT were configured, followed by rigorous firewall rules. Finally, the network's security posture was validated using Nmap penetration testing and Wireshark packet analysis.

Phase 1: Network Architecture & Topology

1.1 Subnetting Strategy

To segment the single 192.168.10.0/24 network, Variable Length Subnet Masking (VLSM) was utilized to create smaller /26 subnets. This provided up to 62 usable hosts per network segment while maintaining strict logical separation:

- **LAN (Employee Network):** 192.168.10.0 /26 (Gateway: 192.168.10.1)
- **DMZ (Web Server):** 192.168.10.64 /26 (Gateway: 192.168.10.65)
- **GUEST (Visitor Wi-Fi):** 192.168.10.128 /26 (Gateway: 192.168.10.129)

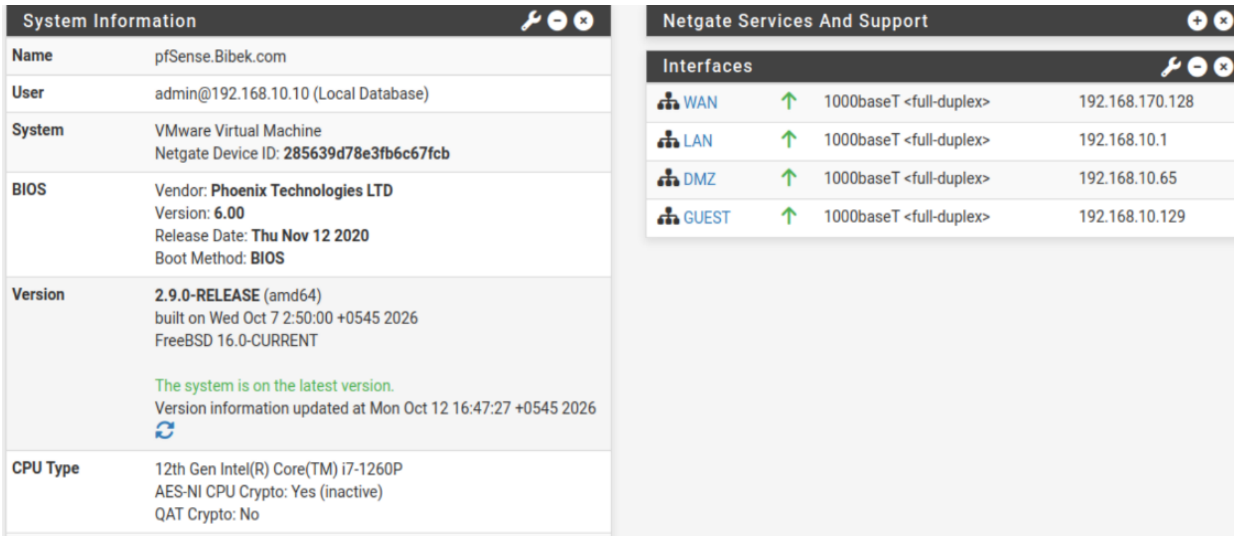


Figure 1: pfSense Interfaces Page

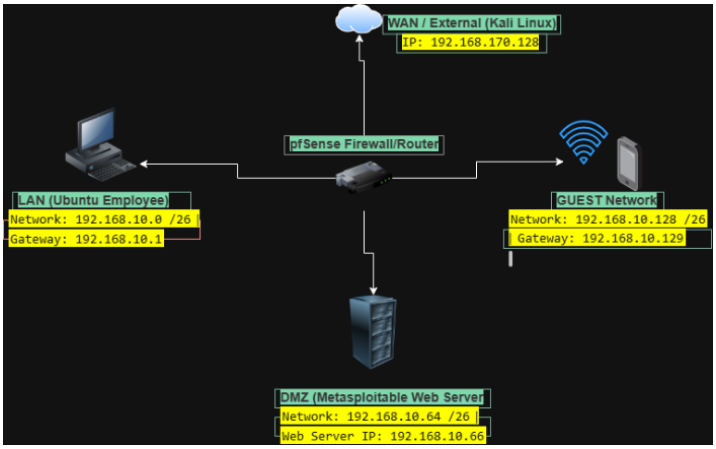


Figure 2: Star Topology Map

Phase 2: Core Services Configuration

To establish network functionality and simulate a real-world enterprise environment, essential routing and naming services were configured on the pfSense router.

2.1 DHCP & DNS Implementation

- **DHCP Server:** Configured on the LAN interface to automatically assign IP addresses, the Default Gateway, and Subnet Masks to internal employee workstations.
- **DNS Resolver:** Configured to translate internal domain queries. The custom domain www.bibek.com was mapped directly to the DMZ Web Server's internal IP (192.168.10.66).

2.2 Network Address Translation (NAT)

To ensure internal host addresses remain hidden from the public internet (WAN), NAT rules were established. Port Forwarding was utilized to allow external traffic to safely reach the internal DMZ web server without exposing the server directly to the internet.

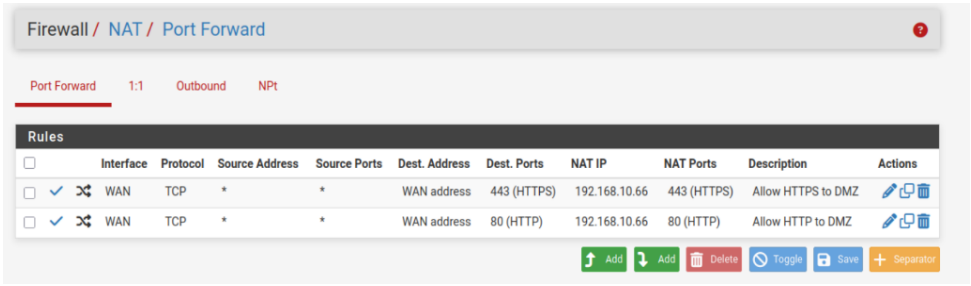


Figure 3: NAT Port Forward Rules

Phase 3: Firewall Rules & Access Control

Strict firewall policies were implemented enforcing a "Default Deny" stance. Traffic is only permitted if it explicitly matches an allow rule.

3.1 Guest Network Isolation

A critical security requirement was ensuring Guest Wi-Fi users could access the internet without compromising the internal company network.

- **Rule 1:** Block all IPv4 traffic originating from GUEST subnets destined for LAN subnets.
- **Rule 2:** Allow all other IPv4 traffic to the internet.

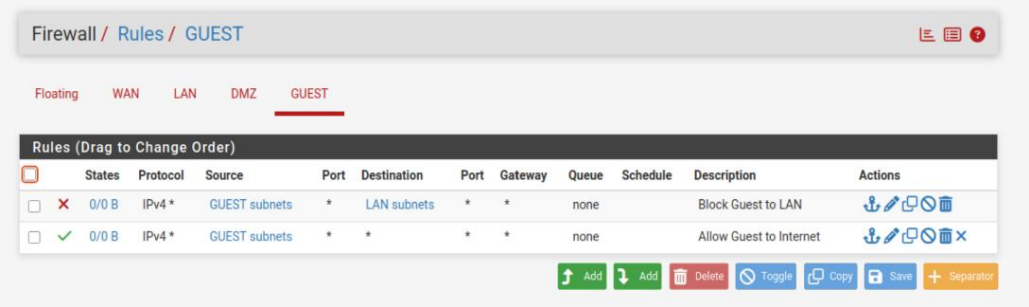


Figure 4: Guest Firewall Rules

3.2 Inbound WAN Security (Anti-Ransomware)

To protect the network from external attacks, specific high-risk ports were explicitly blocked on the WAN interface, while essential web services were allowed.

- **Blocked:** Port 22 (SSH) and Port 445 (SMB) to mitigate brute-force and ransomware vectors.
- **Allowed:** Port 80 (HTTP) and Port 443 (HTTPS) routed to the DMZ web server.

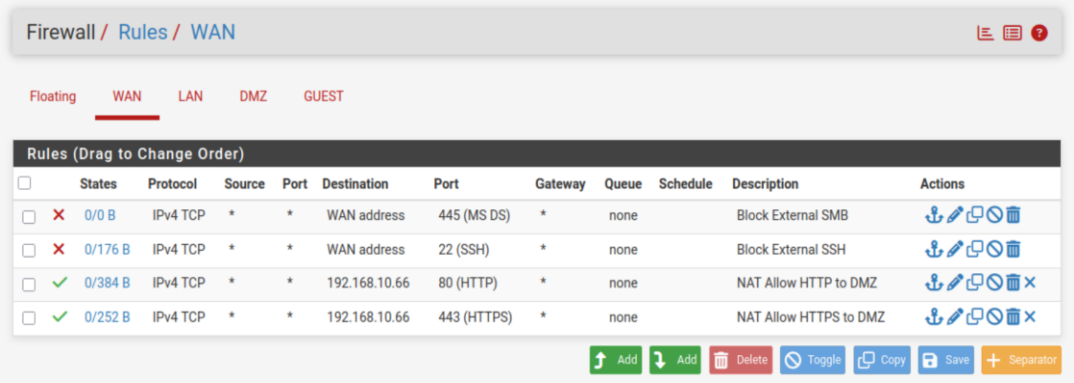


Figure 5: WAN Firewall Rules

Phase 4: Security Auditing & Packet Analysis

To verify the integrity of the firewall rules and network routing, simulated attacks and traffic analyses were conducted.

4.1 External Nmap Penetration Test

An external Kali Linux attacker machine (192.168.170.x) scanned the pfSense WAN IP. The objective was to verify that only web ports were accessible.

- **Results:** The scan successfully confirmed that Port 80 is open and Port 443 is closed (successfully routed through the firewall but no HTTPS service was running on the target). Crucially, Ports 22 and 445 reported as filtered, proving the firewall is silently dropping malicious connection attempts.

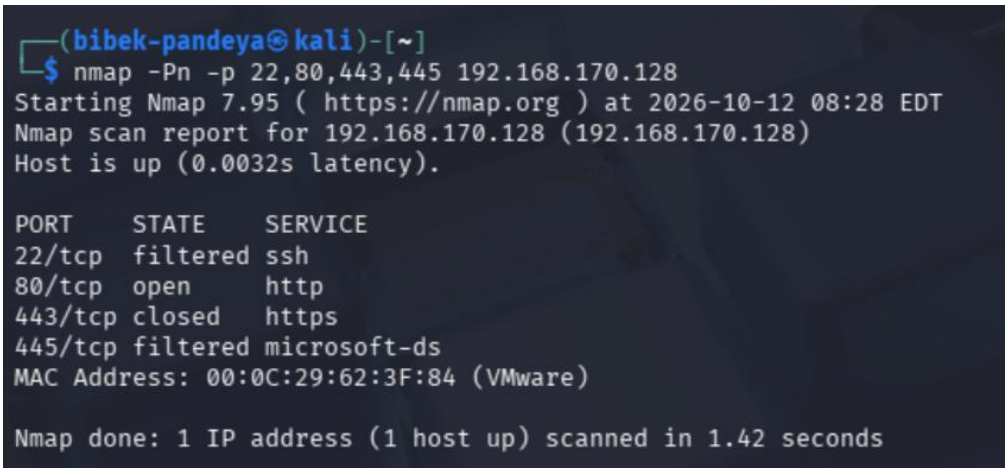


Figure 6: Kali Nmap Scan Results

4.2 Wireshark Deep Packet Inspection

To prove internal connectivity and DNS functionality, a packet capture was initiated on an internal LAN workstation (Ubuntu) while navigating to the web server.

DNS Resolution:
The capture successfully recorded the workstation querying the DNS server for www.bibek.com and receiving the correct internal IP mapping.

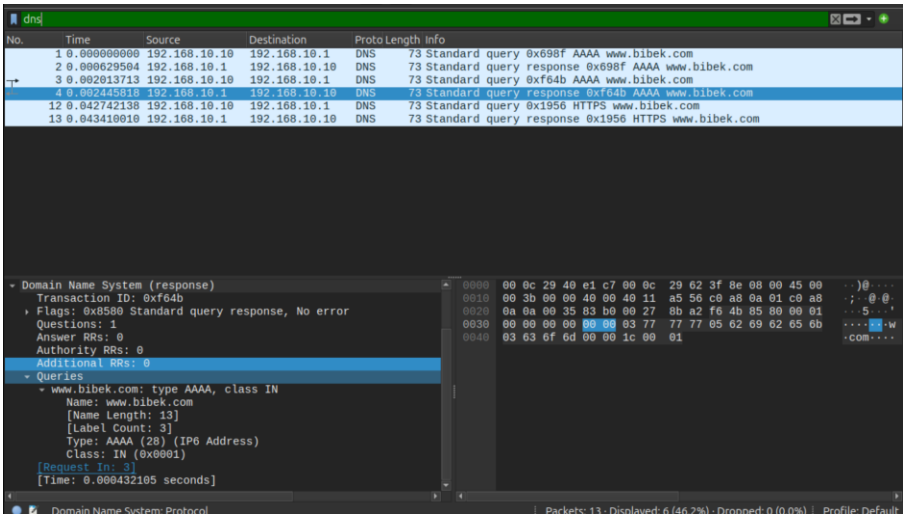


Figure 7: Wireshark DNS Query

TCP 3-Way Handshake:

The capture successfully recorded the foundational TCP connection process required for web traffic, verifying bidirectional communication between the LAN and the DMZ.

1. **[SYN]:** LAN client requests a connection to port 80.
2. **[SYN, ACK]:** DMZ web server acknowledges the request.
3. **[ACK]:** LAN client acknowledges, establishing the session.

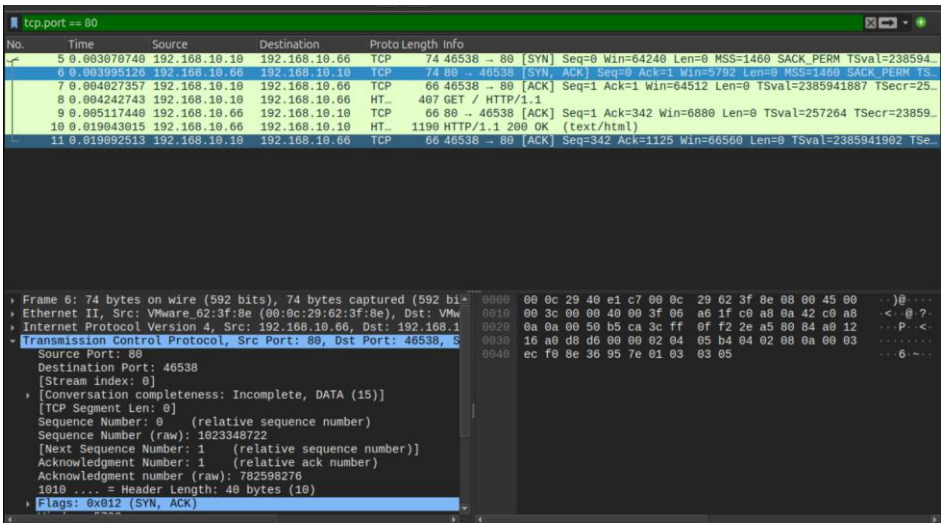


Figure 8: Wireshark TCP Handshake

Thank you